Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

AI Engineer with an MS in Computer Science from USC and 2+ years building **LLM** integrations, **Python** AI services, and **Spark**/Databricks data pipelines, pairing model optimization with analytics that moved marketing and business metrics.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built **AI infrastructure** for an education platform, restructuring async LLM orchestration with FastAPI workers, job-state transitions, and failure isolation to process 1K+ AI jobs at 3x throughput.
- Integrated AI services with internal applications through ETL-style pipelines across PostgreSQL, SQL Server, and cloud storage, running 5K+ monthly jobs at 99%+ reliability for a 500+ user **multi-tenant** platform.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Integrated multimodal **LLMs** (GPT-4o, Claude, Qwen-VL) with OCR into a production grading platform, evaluating 10K+ daily requests for 150+ instructors at 90%+ consistency and 60% faster feedback.
- Optimized model usage with switchable provider routing, prompt chains, and per-call token **cost tracking**, then built concurrent batch pipelines processing 1K+ uploaded files per run with fault tolerance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Trained a GPT-based Decision Transformer in **PyTorch** with AdamW and cosine decay over 100K+ trajectory logs, improving long-horizon planning 18% in simulation through end-to-end data and training pipelines.
- Designed a spatial-temporal attention encoder fusing map geometry with agent trajectories and benchmarked it against three baseline encoders with Hydra and W&B tracking, improving closed-loop **planning stability** 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Used Google **Analytics** on traffic sources, feature usage, and drop-off points to guide marketing and product changes, increasing lead conversions 15%, cutting resource wastage 42%, and expanding market reach.
- Built a MERN resource-management platform with eight microservices for lead management, billing, and resource allocation, synchronizing 2M+ records across multiple locations through REST APIs and real-time **WebSockets**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built hourly **Databricks** and Spark pipelines over 50K+ booking and payment events, aggregating peak booking hours, most-used services, daily active users, and revenue for internal dashboards and client reporting.
- Developed **Flask APIs** for bookings, time-slot pricing, and coupon validation with Firebase real-time sync, sustaining sub-200ms latency and 40% faster booking confirmations for a cross-platform home-services booking app.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
AI/ML: LLM Integration, Prompt Engineering, RAG, Embeddings, PyTorch, Transformers, Model Optimization, Fine-Tuning, Feature Engineering, Hydra, W&B
Data Processing: Databricks, Apache Spark, ETL Pipelines, PostgreSQL, MongoDB, Firebase
AI Infrastructure: FastAPI, Flask, REST APIs, Async Job Orchestration, Microservices, Docker, AWS, Azure, GCP
Analytics & Product: Google Analytics, Funnel Analysis, Recommendation Systems, Dashboards, React
Practices: Rapid Prototyping, CI/CD, Agile, Cross-functional Collaboration

Projects

Midterm Rental Pricing and Listing Copilot | *Python, PySpark, Databricks, FastAPI, Claude, PostgreSQL*
- Designed a **PySpark** pipeline over public rental listings and rent indexes feeding a demand-forecasting model for monthly price recommendations, plus an **LLM agent** drafting listing copy and campaign variants via FastAPI.

Big-Brain Live | *LLaMA-2, FAISS, Embeddings, PostgreSQL, GCP*
- Built an LLM document platform using LLaMA-2, transformer **embeddings**, and FAISS over 50K+ documents, with an incremental ingestion pipeline that cut retrieval latency 40% at 95% query accuracy.

Trend Tracker with Fashion Recommendation System Live | *CNN, React.js, Material UI, Strapi, AWS EC2*
- Developed an e-commerce trend tracker with a CNN-based **recommendation system** on React and AWS EC2, improving recommendation accuracy from 78% to 92% and lifting engagement across 3M+ users.